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River Eelgrass Detection using Very High-Resolution Satellite Imagery

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***Abstract*— River eelgrass has a vital role in the coastal ecosystem, which can be threatened by human activities and climate change. Accurate mapping of eelgrass habitats is important for sustainable eelgrass conservation and management. A large portion of the Shediac river in the province of New Brunswick, Canada is covered by eelgrass, which can be threatened by different natural and anthropogenic activities, such as bridge construction. Thus, mapping and monitoring eelgrass in this river are of interest to the province. In this study, a very high-resolution Worldview-3 image along with an Iterative Self-Organizing Data Analysis Technique (ISODATA) unsupervised classification algorithm was applied to map eelgrass over a portion of Shediac river. The results indicated a reasonable classification accuracy, and it was observed that 90,416.43 m^2 of the study area was covered by river eelgrass.**

Keywords*— *Aquatic vegetation, Worldview-3, Machine learning, Remote Sensing, Classification

I. Introduction

River eelgrass plays an important role in inland water environments. This habitat provides food and shelter for various marine species, produces oxygen, improves water quality, reduces coastal erosion, and stores greenhouse gases. Eelgrass usually lives on the low shore to a depth of 5-meter. It usually has a height of 0.2-1.5 meters and grows on sand, mud, and pebble bottoms of subtidal zones [1].

It is important to monitor eelgrass environments using advanced techniques, such as Remote Sensing (RS) methods. In this regard, various satellites provide suitable spatio-temporal datasets of the eelgrass. RS offers valuable tools to monitor and map almost all marine habitats by decreasing field survey risks and supplying high spatial/temporal resolutions. Fusing various types of RS datasets also enables filling a few gaps of data limitation and weather/time-based obstacles.

Generally, four types of RS datasets can be used for eelgrass mapping. These are multi-spectral satellite and drone imagery [2], [3], Synthetic Aperture Radar (SAR) imagery [4], Sound Navigation and Ranging (SONAR) data [5], and Light Detection and Ranging (LiDAR) point cloud data [1]. Considering the pros and cons of these datasets, multi-spectral satellites images have widely been utilized for eelgrass mapping. For example, [2] investigated and compared the efficacy of high spatial resolution airborne and satellite (IKONOS) imagery for mapping eelgrass distribution at Sidney Spit, Gulf Islands National Park Reserve of Canada using maximum likelihood classification. It was reported that atmospheric correction and deep-water masking can highly increase the accuracy of the results. Landsat images have also been used to monitor eelgrass habitats [6], [7]. For example, [8] triggered change detection-based techniques to monitor eelgrass using Landsat data. Later, [9]–[11] provided advanced mapping of eelgrass habitats. Moreover, [4] recently developed a method based on a combination of SAR and multi-spectral satellite imagery along with machine learning algorithms (e.g., random forest and extreme gradient boosting). They also showed how open-source RS data and open-source software packages could prepare fast and reliable estimation of seagrass ecosystems.

Additionally, machine learning models have extensively been applied to map various habitats, including eelgrass. Various studies have discussed machine learning models and their benefits and limitations to classify aquatic habitats using RS images [12]–[14]. Although supervised classification algorithms usually result in a better accuracy compared to unsupervised models, they require in-situ data. Thus, there are many studies which have used unsupervised algorithms [15], such as Iterative Self-Organizing Data Analysis Technique (ISODATA) [16], [17], K-Means clustering [18], [19], and maximum likelihood [20], [21].

Considering the advantages of very high-resolution multi-spectral satellite images in accurate mapping of various aquatic vegetations, we mapped eelgrass habitats using Worldview-3 images with a spatial resolution of 30cm. To this end, an unsupervised classification algorithm was implemented.

The remainder of the paper is structured as follows: Section 2 describes the study area and provides the characteristics of satellite data. Section 3 presents the methodology, including the preprocessing steps of satellite images and the architecture of the machine learning classifier. In Section 4, we examined the proposed methodology and discussed the obtained results. Finally, the main conclusions are provided in Section 5.

II. STUDY AREA AND DATA

A. Study Area

The study area is a portion of the Shediac river (Fig.1), located in south-eastern New Brunswick, Canada, and originated from Scoudouc to Lutes Mountain. Shediac river is one of the two major rivers, which empty into the Shediac Bay, a vital watershed in Canada with important coastal and natural features. This ecosystem has been endangered by various human activities, such as the construction of bridges.

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Fig. 1. The location of study aera: Shediac River.

B. *Satellite Data*

In this study, a Worldview-3 (TABLE I) multi-spectral image, captured on June 1, 2018, was utilized to map eelgrass in the Shediac river. In addition to panchromatic, visible, and shortwave infrared bands, Worldview-3 has also the coastal blue, yellow, red edge, and near-infrared bands to improve aquatic vegetation mapping. The spatial resolutions of the panchromatic and other multi-spectral bands were 0.3 and 1.24 meters at nadir, respectively. In this study, all of these spectral bands were applied to a machine learning algorithm to obtain the highest possible classification accuracy.

TABLE I. Worldview-3 Satellite Characteristics.

Launch Date	Revisit Frequency	Sensor	Bands	Spatial Resolution (m)
2014	< 1 day	WV-3 PAN	Panchromatic	0.31
		WV-3 MSS	Multispectral	1.24
		WV-3 SWIR	Shortwave infrared	3.7
		WV-3 CAVIS	CAVIS*	30

*CAVIS: Cloud, Aerosol, water Vapor, Ice, Snow

III. METHODOLOGY

The radiometric and geometric accuracies of the satellite image was initially investigated. Then, the pansharpening algorithm was applied to produce an image with a spatial resolution of 30 cm for all spectral bands. Subsequently, the pansharpened image was masked over deep-water areas to produce an image over shallow water regions. This produced image was the input for the machine learning algorithm.

In this study, an Iterative Self-Organizing Data Analysis Technique (ISODATA) unsupervised classification algorithm was implemented to identify eelgrass from non-eelgrass due to the lack of in-situ observations. ISODATA iteratively assigns pixels to clusters based on the shortest distances to the center of them using the averaging of these clusters. In fact, the mean of each class is iteratively calculated to reach a final convergence [16], [17]. Therefore, the final classification result had two groups of pixels: Eelgrass and Non-eelgrass.

IV. RESULTS

A binary classification was performed using the ISODATA unsupervised classification algorithm to distinguish eelgrass from non-eelgrass. Fig. 2a shows the produced map of eelgrass over the Shediac river. A zoomed image was also provided to see the accuracy of the map (see Fig. 2b). Based on the results, the classified map had a high correlation with the very high-resolution satellite image (Fig. 2c) in terms of the locations of eelgrass. Additionally, the propagation of two classes in Fig. 2b demonstrates the visual accuracy of the map, which can be also observed as dark pixels in Fig. 2c. It was also identified that the non-eelgrass areas were mostly sand. Furthermore, it was observed that 90,416.43 m^2 of the study area was covered by eelgrass, which was approximately 20% of the study area. It was also concluded that this region was mostly dominated by non-eelgrass habitats.

It was observed that the unsupervised algorithm used in this study could accurately distinguish regions covered by eelgrass, which decreases the risk and expenses of in-situ measurements. It is worth mentioning that the accuracy and reliability of the produced eelgrass map could be enhanced by other RS data, such as drone images, SAR, and LiDAR imagery in future studies.

V. CONCLUSION

Natural and human activities can negatively affect river habitats, such as eelgrass. Due to the critical impacts of this habitat on ecosystems, it is highly required to develop efficient models for their monitoring. This study showed that very high-resolution satellite images along with machine learning algorithms enabled reliable spatio-temporal mapping of eelgrass. Additionally, advanced machine learning algorithms could improve the accuracy of the produced map. Despite the lack of in-situ data, the visual investigations showed that the produced map using an unsupervised model had a reasonable accuracy in identifying eelgrass. It is suggested to improve the accuracy of the map by (1) acquiring higher resolution drone

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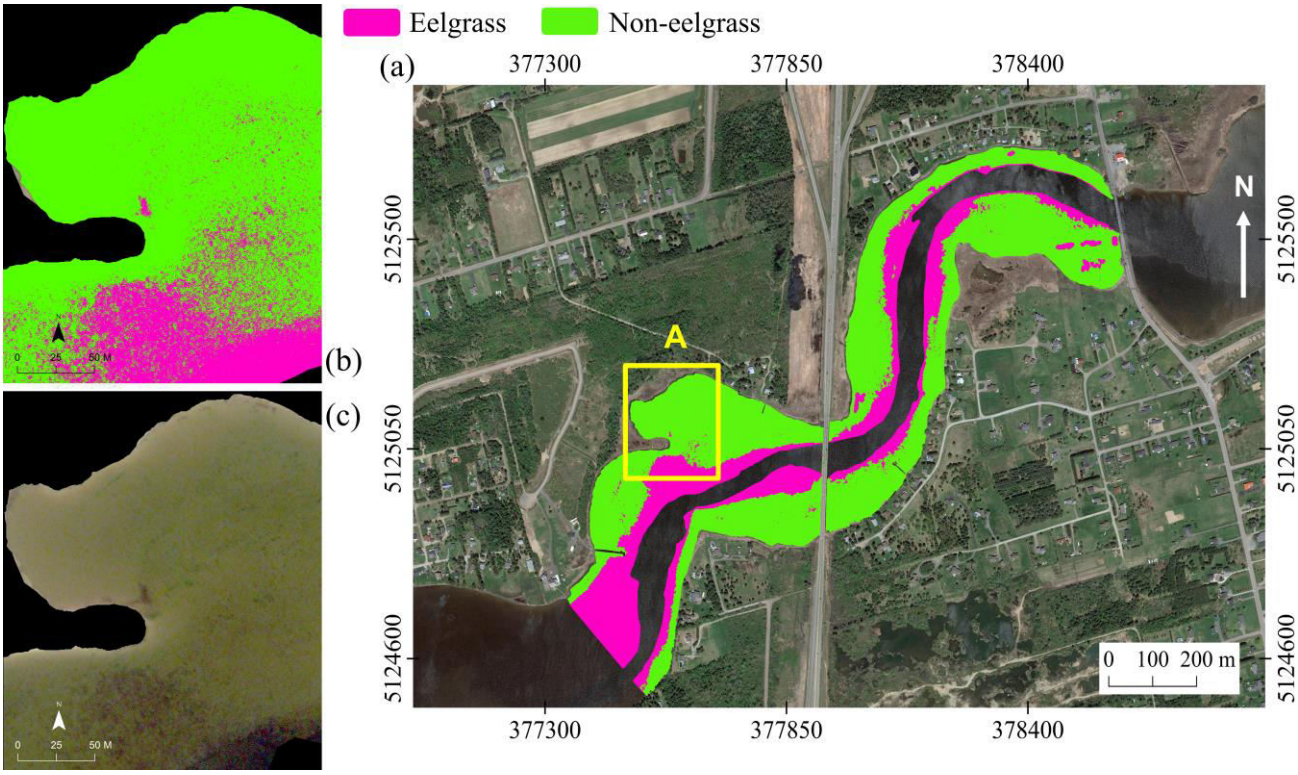

Fig. 2. (a) The eelgrass map of the Shediac river using the ISODATA algorithm and Woldview-3 imagery, along with the (b) a zoomed imaged from the Shediac river (region A in Figure 1a) and the (c) very high-resolution satellite imagery.

images; and (2) implementing more advanced supervised machine learning algorithms, such as object-based random forest models.

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